

# Introduction to Calculus

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# Integration of Rational Functions by Partial Fractions

In this section, we will learn:

How to integrate rational functions  
by reducing them to a sum of simpler fractions.

## PARTIAL FRACTIONS

We show how to integrate any rational function (a ratio of polynomials) by expressing it as a sum of simpler fractions, called partial fractions.

- We already know how to integrate partial functions.

## INTEGRATION BY PARTIAL FRACTIONS

To illustrate the method, observe that, by taking the fractions  $2/(x - 1)$  and  $1/(x - 2)$  to a common denominator, we obtain:

$$\begin{aligned}\frac{2}{x-1} - \frac{1}{x+2} &= \frac{2(x+2) - (x-1)}{(x-1)(x+2)} \\ &= \frac{x+5}{x^2+x-2}\end{aligned}$$

## INTEGRATION BY PARTIAL FRACTIONS

If we now reverse the procedure, we see how to integrate the function on the right side of this equation:

$$\begin{aligned}\int \frac{x+5}{x^2+x-2} dx &= \int \left( \frac{2}{x-1} - \frac{1}{x+2} \right) dx \\ &= 2 \ln |x-1| - \ln |x+2| + C\end{aligned}$$

## INTEGRATION BY PARTIAL FRACTIONS

To see how the method of partial fractions works in general, let's consider a rational function

$$f(x) = \frac{P(x)}{Q(x)}$$

where  $P$  and  $Q$  are polynomials.

## PROPER FUNCTION

It's possible to express  $f$  as a sum of simpler fractions if the degree of  $P$  is less than the degree of  $Q$ .

Such a rational function is called proper.

## DEGREE OF $P$

Recall that, if

$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$$

where  $a_n \neq 0$ , then the degree of  $P$  is  $n$   
and we write  $\deg(P) = n$ .



## PARTIAL FRACTIONS

If  $f$  is improper, that is,  $\deg(P) \geq \deg(Q)$ , then we must take the preliminary step of dividing  $Q$  into  $P$  (by long division).

- This is done until a remainder  $R(x)$  is obtained such that  $\deg(R) < \deg(Q)$ .

The division statement is

$$f(x) = \frac{P(x)}{Q(x)} = S(x) + \frac{R(x)}{Q(x)}$$

where  $S$  and  $R$  are also polynomials.

## PARTIAL FRACTIONS

As the following example illustrates, sometimes, this preliminary step is all that is required.

## PARTIAL FRACTIONS

### Example 1

Find  $\int \frac{x^3 + x}{x-1} dx$

- The degree of the numerator is greater than that of the denominator.
- So, we first perform the long division.

- This enables us to write:

$$\begin{aligned}\int \frac{x^3 + x}{x-1} dx &= \int \left( x^2 + x + 2 + \frac{2}{x-1} \right) dx \\ &= \frac{x^3}{3} + \frac{x^2}{2} + 2x + 2 \ln |x-1| + C\end{aligned}$$

## PARTIAL FRACTIONS

The next step is to factor the denominator  $Q(x)$  as far as possible.

## FACTORISATION OF $Q(x)$

It can be shown that any polynomial  $Q$  can be factored as a product of:

- Linear factors (of the form  $ax + b$ )
- Irreducible quadratic factors (of the form  $ax^2 + bx + c$ , where  $b^2 - 4ac < 0$ ).

## FACTORISATION OF $Q(x)$

For instance, if  $Q(x) = x^4 - 16$ , we could factor it as:

$$\begin{aligned} Q(x) &= (x^2 - 4)(x^2 + 4) \\ &= (x - 2)(x + 2)(x^2 + 4) \end{aligned}$$



## FACTORISATION OF $Q(x)$

The third step is to express the proper rational function  $R(x)/Q(x)$  as a sum of partial fractions of the form:

$$\frac{A}{(ax + b)^i} \quad \text{or} \quad \frac{Ax + B}{(ax^2 + bx + c)^j}$$

## FACTORISATION OF $Q(x)$

A theorem in algebra guarantees that it is always possible to do this.

- We explain the details for the four cases that occur.

## CASE 1

The denominator  $Q(x)$   
is a product of distinct linear  
factors.

## CASE 1

This means that we can write

$$Q(x) = (a_1x + b_1) (a_2x + b_2) \cdots (a_kx + b_k)$$

where no factor is repeated (and no factor is a constant multiple of another).

## CASE 1

## Equation 2

In this case, the partial fraction theorem states that there exist constants  $A_1, A_2, \dots, A_k$  such that:

$$\frac{R(x)}{Q(x)} = \frac{A_1}{a_1x + b_1} + \frac{A_2}{a_2x + b_2} + \dots + \frac{A_k}{a_kx + b_k}$$

## CASE 1

These constants can be determined as in the following example.

Evaluate  $\int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx$

- The degree of the numerator is less than the degree of the denominator.
- So, we don't need to divide.

We factor the denominator as:

$$\begin{aligned} 2x^3 + 3x^2 - 2x &= x(2x^2 + 3x - 2) \\ &= x(2x - 1)(x + 2) \end{aligned}$$

- It has three distinct linear factors.



So, the partial fraction decomposition of the integrand (Equation 2) has the form

$$\frac{x^2 + 2x - 1}{x(2x - 1)(x + 2)} = \frac{A}{x} + \frac{B}{2x - 1} + \frac{C}{x + 2}$$

## PARTIAL FRACTIONS

E. g. 2—Equation 4

To determine the values of  $A$ ,  $B$ , and  $C$ , we multiply both sides of the equation by the product of the denominators,  $x(2x - 1)(x + 2)$ , obtaining:

$$x^2 + 2x + 1 = A(2x - 1)(x + 2) + Bx(x + 2) + Cx(2x - 1)$$

## PARTIAL FRACTIONS

E. g. 2—Equation 5

Expanding the right side of Equation 4 and writing it in the standard form for polynomials, we get:

$$\begin{aligned}x^2 + 2x + 1 &= (2A + B + 2C)x^2 \\ &\quad + (3A + 2B - C)x - 2A\end{aligned}$$

The polynomials in Equation 5 are identical.  
So, their coefficients must be equal.

- The coefficient of  $x^2$  on the right side,  $2A + B + 2C$ , must equal that of  $x^2$  on the left side—namely, 1.
- Likewise, the coefficients of  $x$  are equal and the constant terms are equal.

## PARTIAL FRACTIONS

### Example 2

This gives the following system of equations for  $A$ ,  $B$ , and  $C$ :

$$2A + B + 2C = 1$$

$$3A + 2B - C = 2$$

$$-2A = -1$$

Solving, we get:

- $A = \frac{1}{2}$

- $B = \frac{1}{5}$

- $C = -\frac{1}{10}$

Hence,

$$\begin{aligned} & \int \frac{x^2 + 2x - 1}{2x^3 + 3x^2 - 2x} dx \\ &= \int \left( \frac{1}{2} \frac{1}{x} + \frac{1}{5} \frac{1}{2x-1} - \frac{1}{10} \frac{1}{x+2} \right) dx \\ &= \frac{1}{2} \ln |x| + \frac{1}{10} \ln |2x-1| - \frac{1}{10} \ln |x+2| + K \end{aligned}$$

In integrating the middle term,  
we have made the mental substitution  
 $u = 2x - 1$ , which gives  
 $du = 2 \, dx$  and  $dx = du/2$ .



## NOTE

We can use an alternative method to find the coefficients  $A$ ,  $B$ , and  $C$  in Example 2.

## NOTE

Equation 4 is an identity.

It is true for every value of  $x$ .

- Let's choose values of  $x$  that simplify the equation.

## NOTE

If we put  $x = 0$  in Equation 4, the second and third terms on the right side vanish, and the equation becomes  $-2A = -1$ .

- Hence,  $A = \frac{1}{2}$ .

## NOTE

Likewise,  $x = \frac{1}{2}$  gives  $5B/4 = 1/4$   
and  $x = -2$  gives  $10C = -1$ .

- Hence,  $B = 1/5$  and  $C = -1/10$ .

## NOTE

You may object that Equation 3 is not valid for  $x = 0$ ,  $\frac{1}{2}$ , or  $-2$ .

- So, why should Equation 4 be valid for those values?

## NOTE

In fact, Equation 4 is true for all values of  $x$ , even  $x = 0$ ,  $\frac{1}{2}$ , and  $-2$  .

- See Exercise 69 for the reason.

Find  $\int \frac{dx}{x^2 - a^2}$ , where  $a \neq 0$ .

- The method of partial fractions gives:

$$\frac{1}{x^2 - a^2} = \frac{1}{(x - a)(x + a)} = \frac{A}{x - a} + \frac{B}{x + a}$$

- Therefore,  $A(x + a) + B(x - a) = 1$

We use the method of the preceding note.

- We put  $x = a$  in the equation and get  $A(2a) = 1$ .  
So,  $A = 1/(2a)$ .
- If we put  $x = -a$ , we get  $B(-2a) = 1$ .  
So,  $B = -1/(2a)$ .



Therefore,

$$\begin{aligned}\int \frac{dx}{x^2 - a^2} &= \frac{1}{2a} \int \left( \frac{1}{x - a} - \frac{1}{x + a} \right) dx \\ &= \frac{1}{2a} (\ln |x - a| - \ln |x + a|) + C\end{aligned}$$

## PARTIAL FRACTIONS

E. g. 3—Formula 6

Since  $\ln x - \ln y = \ln(x/y)$ , we can write the integral as:

$$\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \ln \left| \frac{x-a}{x+a} \right| + C$$

- See Exercises 55–56 for ways of using Formula 6.

## CASE 2

$Q(x)$  is a product of linear factors, some of which are repeated.

## CASE 2

Suppose the first linear factor  $(a_1x + b_1)$  is repeated  $r$  times.

- That is,  $(a_1x + b_1)^r$  occurs in the factorization of  $Q(x)$ .

## CASE 2

## Equation 7

Then, instead of the single term  $A_1/(a_1x + b_1)$  in Equation 2, we would use:

$$\frac{A_1}{a_1x + b_1} + \frac{A_2}{(a_1x + b_1)^2} + \dots + \frac{A_r}{(a_1x + b_1)^r}$$

## CASE 2

By way of illustration, we could write:

$$\frac{x^3 - x + 1}{x^2(x-1)^3} = \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-1} + \frac{D}{(x-1)^2} + \frac{E}{(x-1)^3}$$

- However, we prefer to work out in detail a simpler example, as follows.

## PARTIAL FRACTIONS

### Example 4

Find  $\int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx$

- The first step is to divide.

- The result of long division is: 
$$\frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1}$$
$$= x + 1 + \frac{4x}{x^3 - x^2 - x + 1}$$

The second step is to factor the denominator  $Q(x) = x^3 - x^2 - x + 1$ .

- Since  $Q(1) = 0$ , we know that  $x - 1$  is a factor, and we obtain:

$$\begin{aligned}x^3 - x^2 - x + 1 &= (x - 1)(x^2 - 1) \\&= (x - 1)(x - 1)(x + 1) \\&= (x - 1)^2(x + 1)\end{aligned}$$



The linear factor  $x - 1$  occurs twice.

So, the partial fraction decomposition is:

$$\frac{4x}{(x-1)^2(x+1)} = \frac{A}{x-1} + \frac{B}{(x-1)^2} + \frac{C}{x+1}$$

Multiplying by the least common denominator,  $(x - 1)^2 (x + 1)$ , we get:

$$\begin{aligned} 4x &= A(x - 1)(x + 1) + B(x + 1) + C(x - 1)^2 \\ &= (A + C)x^2 + (B - 2C)x + (-A + B + C) \end{aligned}$$

Now, we equate coefficients:

$$A + C = 0$$

$$B - 2C = 4$$

$$-A + B + C = 0$$

Solving, we obtain:

$$A = 1$$

$$B = 2$$

$$C = -1$$

Thus,

$$\begin{aligned} & \int \frac{x^4 - 2x^2 + 4x + 1}{x^3 - x^2 - x + 1} dx \\ &= \int \left[ x + 1 + \frac{1}{x-1} + \frac{2}{(x-1)^2} - \frac{1}{x+1} \right] dx \\ &= \frac{x^2}{2} + x + \ln |x-1| - \frac{2}{x-1} - \ln |x+1| + K \\ &= \frac{x^2}{2} + x - \frac{2}{x-1} + \ln \left| \frac{x-1}{x+1} \right| + K \end{aligned}$$

### CASE 3

$Q(x)$  contains irreducible quadratic factors, none of which is repeated.

### CASE 3

### Formula 9

If  $Q(x)$  has the factor  $ax^2 + bx + c$ , where  $b^2 - 4ac < 0$ , then, in addition to the partial fractions in Equations 2 and 7, the expression for  $R(x)/Q(x)$  will have a term of the form

$$\frac{Ax + B}{ax^2 + bx + c}$$

where  $A$  and  $B$  are constants to be determined.

### CASE 3

For instance, the function given by  $f(x) = x/[(x - 2)(x^2 + 1)(x^2 + 4)]$  has a partial fraction decomposition of the form

$$\frac{x}{(x - 2)(x^2 + 1)(x^2 + 4)} = \frac{A}{x - 2} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{x^2 + 4}$$



### CASE 3

### Formula 10

The term in Formula 9 can be integrated by completing the square and using the formula

$$\int \frac{dx}{x^2 + a^2} = \frac{1}{a} \tan^{-1} \left( \frac{x}{a} \right) + C$$

## PARTIAL FRACTIONS

### Example 5

Evaluate  $\int \frac{2x^2 - x + 4}{x^3 + 4x} dx$

- As  $x^3 + 4x = x(x^2 + 4)$  can't be factored further, we write:

$$\frac{2x^2 - x + 4}{x(x^2 + 4)} = \frac{A}{x} + \frac{Bx + C}{x^2 + 4}$$

Multiplying by  $x(x^2 + 4)$ , we have:

$$\begin{aligned} 2x^2 - x + 4 &= A(x^2 + 4) + (Bx + C)x \\ &= (A + B)x^2 + Cx + 4A \end{aligned}$$

Equating coefficients, we obtain:

$$A + B = 2 \quad C = -1 \quad 4A = 4$$

- Thus,  $A = 1$ ,  $B = 1$ , and  $C = -1$ .

Hence,

$$\int \frac{2x^2 - x + 4}{x^3 + 4x} dx = \int \left( \frac{1}{x} + \frac{x-1}{x^2 + 4} \right) dx$$

In order to integrate the second term, we split it into two parts:

$$\int \frac{x-1}{x^2+4} dx = \int \frac{x}{x^2+4} dx - \int \frac{1}{x^2+4} dx$$

We make the substitution  $u = x^2 + 4$  in the first of these integrals so that  $du = 2x \, dx$ .

We evaluate the second integral by means of Formula 10 with  $a = 2$ :

$$\begin{aligned} & \int \frac{2x^2 - x + 4}{x(x^2 + 4)} dx \\ &= \int \frac{1}{x} dx + \int \frac{x}{x^2 + 4} dx - \int \frac{1}{x^2 + 4} dx \\ &= \ln |x| + \frac{1}{2} \ln(x^2 + 4) - \frac{1}{2} \tan^{-1}(x/2) + K \end{aligned}$$



## PARTIAL FRACTIONS

### Example 6

Evaluate  $\int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx$

- The degree of the numerator is not less than the degree of the denominator.

- So, we first divide and obtain: 
$$\frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} = 1 + \frac{x - 1}{4x^2 - 4x + 3}$$

## PARTIAL FRACTIONS

### Example 6

Notice that the quadratic  $4x^2 - 4x + 3$  is irreducible because its discriminant is  $b^2 - 4ac = -32 < 0$ .

- This means it can't be factored.
- So, we don't need to use the partial fraction technique.

To integrate the function, we complete the square in the denominator:

$$4x^2 - 4x + 3 = (2x - 1)^2 + 2$$

- This suggests we make the substitution  $u = 2x - 1$ .
- Then,  $du = 2 \, dx$ , and  $x = \frac{1}{2}(u + 1)$ .

Thus,

$$\begin{aligned} & \int \frac{4x^2 - 3x + 2}{4x^2 - 4x + 3} dx \\ &= \int \left( 1 + \frac{x-1}{4x^2 - 4x + 3} \right) dx \\ &= x + \frac{1}{2} \int \frac{\frac{1}{2}(u+1) - 1}{u^2 + 2} du \\ &= x + \frac{1}{4} \int \frac{u-1}{u^2 + 2} du \end{aligned}$$

## PARTIAL FRACTIONS

### Example 6

$$= x + \frac{1}{4} \int \frac{u}{u^2 + 2} du - \frac{1}{4} \int \frac{1}{u^2 + 2} du$$

$$= x + \frac{1}{8} \ln(u^2 + 2) - \frac{1}{4} \cdot \frac{1}{\sqrt{2}} \tan^{-1} \left( \frac{u}{\sqrt{2}} \right) + C$$

$$= x + \frac{1}{8} \ln(4x^2 - 4x + 3) - \frac{1}{4\sqrt{2}} \tan^{-1} \left( \frac{2x-1}{\sqrt{2}} \right) + C$$

## NOTE

Example 6 illustrates the general procedure for integrating a partial fraction of the form

$$\frac{Ax + B}{ax^2 + bx + c} \quad \text{where } b^2 - 4ac < 0$$

## NOTE

We complete the square in the denominator and then make a substitution that brings the integral into the form

$$\int \frac{Cu + D}{u^2 + a^2} du = C \int \frac{u}{u^2 + a^2} du + D \int \frac{1}{u^2 + a^2} du$$

- Then, the first integral is a logarithm and the second is expressed in terms of  $\tan^{-1}$ .

## CASE 4

$Q(x)$  contains  
a repeated irreducible  
quadratic factor.



## CASE 4

Suppose  $Q(x)$  has the factor

$$(ax^2 + bx + c)^r$$

where  $b^2 - 4ac < 0$ .

## CASE 4

## Formula 11

Then, instead of the single partial fraction (Formula 9), the sum

$$\frac{A_1x + B_1}{ax^2 + bx + c} + \frac{A_2x + B_2}{(ax^2 + bx + c)^2} + \cdots + \frac{A_rx + B_r}{(ax^2 + bx + c)^r}$$

occurs in the partial fraction decomposition of  $R(x)/Q(x)$ .

## CASE 4

Each of the terms in Formula 11  
can be integrated by first completing  
the square.

Write out the form of the partial fraction decomposition of the function

$$\frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3}$$

We have:

$$\begin{aligned} & \frac{x^3 + x^2 + 1}{x(x-1)(x^2 + x + 1)(x^2 + 1)^3} \\ &= \frac{A}{x} + \frac{B}{x-1} + \frac{Cx + D}{x^2 + x + 1} + \frac{Ex + F}{x^2 + 1} \\ & \quad + \frac{Gx + h}{(x^2 + 1)^2} + \frac{Ix + J}{(x^2 + 1)^3} \end{aligned}$$

## PARTIAL FRACTIONS

### Example 8

Evaluate  $\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$

- The form of the partial fraction decomposition is:

$$\frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} = \frac{A}{x} + \frac{Bx + C}{x^2 + 1} + \frac{Dx + E}{(x^2 + 1)^2}$$

Multiplying by  $x(x^2 + 1)^2$ ,  
we have:

$$\begin{aligned} & -x^3 + 2x^2 - x + 1 \\ &= A(x^2 + 1)^2 + (Bx + C)x(x^2 + 1) + (Dx + E)x \\ &= A(x^4 + 2x^2 + 1) + B(x^4 + x^2) + C(x^3 + x) + Dx^2 + Ex \\ &= (A + B)x^4 + Cx^3 + (2A + B + D)x^2 + (C + E)x + A \end{aligned}$$

If we equate coefficients,

we get the system

$$A + B = 0$$

$$C = -1$$

$$2A + B + D = 2$$

$$C + E = -1$$

$$A = 1$$

- This has the solution

$$A = 1, B = -1, C = -1, D = 1, E = 0.$$



Thus,

$$\int \frac{1 - x + 2x^2 - x^3}{x(x^2 + 1)^2} dx$$

$$= \int \left( \frac{1}{x} - \frac{x+1}{x^2+1} + \frac{x}{(x^2+1)^2} \right) dx$$

$$= \int \frac{dx}{x} - \int \frac{x}{x^2+1} dx - \int \frac{dx}{x^2+1} + \int \frac{x dx}{(x^2+1)^2}$$

$$= \ln |x| - \frac{1}{2} \ln(x^2 + 1) - \tan^{-1} x - \frac{1}{2(x^2 + 1)^2} + K$$

## AVOIDING PARTIAL FRACTIONS

We note that, sometimes,  
partial fractions can be avoided  
when integrating a rational function.

## AVOIDING PARTIAL FRACTIONS

For instance, the integral

$$\int \frac{x^2 + 1}{x(x^2 + 3)} dx$$

could be evaluated by the method  
of Case 3.

## AVOIDING PARTIAL FRACTIONS

However, it is much easier to observe that, if  $u = x(x^2 + 3) = x^3 + 3x$ , then  $du = (3x^2 + 3) dx$  and so

$$\int \frac{x^2 + 1}{x(x^2 + 3)} dx = \frac{1}{3} \ln |x^3 + 3x| + C$$

## RATIONALIZING SUBSTITUTIONS

Some nonrational functions can be changed into rational functions by means of appropriate substitutions.

- In particular, when an integrand contains an expression of the form  $\sqrt[n]{g(x)}$ , then the substitution  $u = \sqrt[n]{g(x)}$  may be effective.

## RATIONALIZING SUBSTITUTIONS Example 9

Evaluate  $\int \frac{\sqrt{x+4}}{x} dx$

- Let  $u = \sqrt{x+4}$
- Then,  $u^2 = x + 4$
- So,  $x = u^2 - 4$  and  $dx = 2u \, du$

## RATIONALIZING SUBSTITUTIONS Example 9

■ Therefore,

$$\begin{aligned}\int \frac{\sqrt{x+4}}{x} dx &= \int \frac{u}{u^2 - 4} 2u du \\ &= 2 \int \frac{u^2}{u^2 - 4} du \\ &= 2 \int \left( 1 + \frac{4}{u^2} - 4 \right) du\end{aligned}$$

## RATIONALIZING SUBSTITUTIONS Example 9

We can evaluate this integral  
by factoring  $u^2 - 4$  as  $(u - 2)(u + 2)$   
and using partial fractions.



## RATIONALIZING SUBSTITUTIONS Example 9

Alternatively, we can use Formula 6

with  $a = 2$ :  $\int \frac{\sqrt{x+4}}{x} dx$

$$= 2 \int du + 8 \int \frac{du}{u^2 - 4}$$

$$= 2u + 8 \cdot \frac{1}{2 \cdot 2} \ln \left| \frac{u-2}{u+2} \right| + C$$

$$= 2\sqrt{x+4} + 2 \ln \left| \frac{\sqrt{x+4}-2}{\sqrt{x+4}+2} \right| + C$$